

Curriculum Vitae

Yue Ma — Experimental Nuclear Physicist

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1. Profile

Experimental nuclear physicist and Senior Research Scientist (permanent) at RIKEN, with twenty years of front-line research and leadership at major international facilities in Japan (J-PARC, SPring-8, KEK) and Germany (GSI, DESY), and a long record of training young researchers and building international collaborations. **Currently experiment spokesperson of J-PARC E73 (hypertriton lifetime), and proposal spokesperson of SPring-8 LEPS2 (nucleon short-range correlations) and the J-PARC antideuteron experiment.**

2. Academic Leadership & Appointments

- J-PARC E73, **hypertriton lifetime — Experiment spokesperson.** Beam time completed; the ${}^4_{\Lambda}\text{H}$ lifetime and ${}^3,4_{\Lambda}\text{H}$ production cross sections are published; the ${}^3_{\Lambda}\text{H}$ lifetime analysis is ongoing.
- SPring-8 LEPS2, **nucleon short-range correlations — Proposal spokesperson.** Probing short-range correlations between nucleons with the Λ hyperon as a probe (ongoing).
- J-PARC antideuteron experiment — **Proposal spokesperson.** Antideuteron–nucleus interaction and annihilation mechanism (ongoing).

3. Signature Achievements

1. **Precise lifetime measurement of the ${}^4_{\Lambda}\text{H}$ hypernucleus** — as spokesperson and corresponding author, proposed the key idea of tagging the fast π^0 with a single high-energy γ and led the measurement, $\tau = 206 \pm 8 \pm 12$ ps (J-PARC E73). Phys. Lett. B **845** (2023); **873** (2026).
2. **First observation of a $\Lambda\pi$ structure near the $\bar{K}N(I=1)$ threshold** — led the Belle data analysis as first author. Phys. Rev. Lett. **130** (2023).
3. **Discovery of the K^-pp ($\bar{K}NN$) kaonic nuclear bound state** — as a founding member, developed the multi-hit TDC and extracted the signal by partial-wave analysis, giving the first evidence for a deeply bound K^-pp state (J-PARC E15). Phys. Lett. B **789** (2019); Phys. Rev. C **102** (2020).
4. **First application of TES microcalorimeters to hadronic-atom spectroscopy** — implemented the SDD data analysis and energy calibration at sub-eV precision (J-PARC E62). Prog. Theor. Exp. Phys. **2016**, 091D01; Phys. Rev. Lett. **128** (2022).
5. **First proposal and demonstration of energy-resolved muon spin spectroscopy (μSR)** — as first and corresponding author, introduced event-by-event positron-energy selection with lead-glass Cherenkov calorimeters and led the demonstration, a $\sim 30\%$ figure-of-merit gain at no cost in beam time (J-PARC MLF). Nucl. Instrum. Meth. A **1092**, 171842, Short communication (2026).

6. **Determination of hard two-photon exchange (OLYMPUS) with the PbF₂ luminosity monitor** — led the development of the luminosity monitor, a PbF₂ Cherenkov calorimeter based on Møller/Bhabha scattering, whose precision underpinned the per-mille e^+p/e^-p elastic cross-section ratio; OLYMPUS found the hard two-photon contribution significantly weaker than theoretical expectations (DESY). Phys. Rev. Lett. **118** (2017); Nucl. Instrum. Meth. A **826** (2016).

4. Education

2006–2009	Doctor of Science, Tohoku University (Sendai)
2004–2006	M.S. in Physical Science, Tohoku University (Sendai)
1998–2002	B.S. in Nuclear Physics, Lanzhou University

5. Appointments

2012–now	Senior Research Scientist (permanent), RIKEN
2009–2012	Postdoctoral Researcher, Helmholtz Institute Mainz, Germany
2002–2004	Research Trainee, Institute of Modern Physics, CAS

6. Grants and Fellowships

2026–2027	Grant-in-Aid for Transformative Research Areas (A), Publicly Offered Research: search for a Σ -hypertriton ($^3_\Sigma\text{H}$) bound state at J-PARC (within E. Hiyama's research area). PI
2024–2028	Grant-in-Aid for Scientific Research (A): ρ -meson mass modification in nuclei with photon beams. Co-investigator
2023–2025	RIKEN Incentive Research Project: exploring matter–antimatter interactions. PI
2017–2020	JSPS Grant-in-Aid for Young Scientists (A) 17H04842: solving the hypertriton lifetime puzzle. PI

7. Student Supervision

2017–2020	C. Han (PhD), Univ. of Chinese Academy of Sciences
2012–2014	Q. Zhang (PhD), Lanzhou University
2010–2011	D. Lin (PhD); Christina, Pascal (M.S.), Univ. of Mainz

8. Service & Invited Talks

2019–2021	Member, RIKEN Scientists' Assembly Steering Committee; coordinator / chief coordinator, RIKEN Incentive Research Fund (Physics)
2022	Invited talk, 14th Int. Conf. on Hypernuclear & Strange Particle Physics (HYP 2022, Prague)
2025	Invited talk, 15th HYP (HYP 2025, Tokyo)

9. Teaching & Languages

2018/23/24	Invited lecturer, Hadron Physics summer course, Univ. of Chinese Academy of Sciences
Languages	Chinese (native) · English (fluent) · Japanese (fluent)

10. Selected Publications (ordered by signature achievement; authorship indicated)

More than **120 publications** (including Belle / Belle II collaboration papers, see the final section); the following are first-author / corresponding-author / spokesperson or otherwise leading contributions.

Note: hadron-physics papers are usually listed alphabetically by surname; the **first author** / **corresponding author** / **spokesperson** tags below mark the actual leading role.

1. T. Akaishi *et al.*, *Precise lifetime measurement of ${}^4_\Lambda H$ hypernucleus using in-flight ${}^4\text{He}(K^-, \pi^0)$ reaction*, Phys. Lett. B **845**, 138128 (2023). [**Spokesperson & corresponding author**]
2. T. Akaishi *et al.*, *Measurement of ${}^3,4\text{He}(K^-, \pi^0){}^3,4_\Lambda H$ reaction cross section and evaluation of hypertriton Λ binding energy*, Phys. Lett. B **873**, 140163 (2026). [**Spokesperson**]
3. **Y. Ma** *et al.*, *First observation of $\Lambda\pi^+$ and $\Lambda\pi^-$ signals near the $\bar{K}N(I=1)$ mass threshold in $\Lambda_c^+ \rightarrow \Lambda\pi^+\pi^+\pi^-$ decay*, Phys. Rev. Lett. **130**, 151903 (2023) [Belle]. [**First author**]
4. S. Ajimura *et al.*, *K^-pp , a $\bar{K}$ -meson nuclear bound state, observed in ${}^3\text{He}(K^-, \Lambda p)n$ reactions*, Phys. Lett. B **789**, 620 (2019). [**Major author**]
5. T. Yamaga *et al.*, *Observation of a $\bar{K}NN$ bound state in the ${}^3\text{He}(K^-, \Lambda p)n$ reaction*, Phys. Rev. C **102**, 044002 (2020). [**Major contributor**]
6. T. Hashimoto *et al.*, *Strong-interaction effects in kaonic-helium isotopes at sub-eV precision with X-ray microcalorimeters*, Phys. Rev. Lett. **128**, 112503 (2022). [**Major contributor**]
7. **Y. Ma** *et al.*, *Demonstration of energy-resolved muon spin spectroscopy using a Cherenkov calorimeter*, Nucl. Instrum. Meth. A **1092**, 171842, Short communication (2026). [**First author & corresponding author**]
8. B. S. Henderson *et al.*, *Hard two-photon contribution to elastic lepton-proton scattering determined by the OLYMPUS experiment*, Phys. Rev. Lett. **118**, 092501 (2017). [**Major contributor; led the PbF₂ luminosity monitor**]
9. **Y. Ma** *et al.*, *γ -ray spectroscopy study of ${}^{11}_\Lambda B$ and ${}^{12}_\Lambda C$* , Eur. Phys. J. A **33**, 243 (2007). [**First author**]
10. K. Hosomi *et al.*, *Precise determination of ${}^{12}_\Lambda C$ level structure by γ -ray spectroscopy*, Prog. Theor. Exp. Phys. **2015**, 081D01 (2015). [**Second author**]

11. Belle / Belle II Collaboration Papers

As a member of the Belle / Belle II collaboration since **2023**, co-author of **70+** collaboration papers; representative papers by theme:

I. B -meson CP violation & CKM matrix elements Time-dependent CP violation ($\sin 2\phi_1$, etc.), $|V_{cb}|/|V_{ub}|$, and the CKM angle ϕ_3 .

1. *Measurement of $\sin 2\phi_1$ in $B^0 \rightarrow J/\psi K_S^0$ decays with a graph-neural-network flavor tagger*, Phys. Rev. D **110**, 012001 (2024) [Belle II].
2. *Measurement of CP violation in $B^0 \rightarrow K_S^0 \pi^0$ decays*, Phys. Rev. Lett. **131**, 111803 (2023) [Belle II].
3. *Determination of $|V_{cb}|$ using $\bar{B}^0 \rightarrow D^{*+} \ell^- \bar{\nu}_\ell$ decays*, Phys. Rev. D **108**, 092013 (2023) [Belle II].
4. *Measurement of inclusive $B \rightarrow X_u \ell \nu$ partial branching fractions and $|V_{ub}|$* , Phys. Rev. D **113**, 032004 (2026) [Belle II].
5. *Determination of the CKM angle ϕ_3 from a combination of Belle and Belle II results*, J. High Energy Phys. **10** (2024) 143 [Belle/Belle II].

II. Rare decays & lepton-flavour universality First evidence for $B^+ \rightarrow K^+ \nu \bar{\nu}$, $R(D^*)$ and other LFU tests, and τ lepton-flavour-violation searches.

1. *Evidence for $B^+ \rightarrow K^+ \nu \bar{\nu}$ decays*, Phys. Rev. D **109**, 112006 (2024) [Belle II].
2. *Test of light-lepton universality with a measurement of $R(D^*)$ using hadronic B tagging*, Phys. Rev. D **110**, 072020 (2024) [Belle II].
3. *First measurement of $R(X_{\tau/\ell})$ as an inclusive test of the $b \rightarrow c \tau \nu$ anomaly*, Phys. Rev. Lett. **132**, 211804 (2024) [Belle II].
4. *Tests of light-lepton universality in angular asymmetries of $B^0 \rightarrow D^{*-} \ell \nu$ decays*, Phys. Rev. Lett. **131**,

181801 (2023) [Belle II].

5. *Measurement of the $B^+ \rightarrow \tau^+ \nu_\tau$ branching fraction with a hadronic tagging method*, Phys. Rev. D **112**, 072002 (2025) [Belle II].
6. *First search for $B \rightarrow X_s \nu \bar{\nu}$ decays*, Phys. Rev. Lett. **136**, 231801 (2026) [Belle II].
7. *Search for lepton-flavor-violating τ decays to $\ell\alpha$* , J. High Energy Phys. **08** (2025) 155 [Belle].
8. *Search for a $\tau^+ \tau^-$ resonance in $e^+ e^- \rightarrow \mu^+ \mu^- \tau^+ \tau^-$ events*, Phys. Rev. Lett. **131**, 121802 (2023) [Belle II].

III. Charm/strange baryon spectroscopy & exotic hadrons(closely aligned with my own research) Ξ_c decays and CP, the first observation of $D_{s0}^*(2317)$ radiative decay, P_{cs} pentaquarks, and $\Xi^0 p / \Omega^- p / \Omega^- n$ dibaryon searches.

1. *Observation of the radiative decay $D_{s0}^*(2317)^+ \rightarrow D_s^{*+} \gamma$* , Phys. Rev. Lett. **136**, 241901 (2026) [Belle/Belle II].
2. *Search for $P_{cs}(4459)$ and $P_{cs}(4338)$ in $\Upsilon(1S, 2S)$ inclusive decays*, Phys. Rev. Lett. **135**, 041901 (2025) [Belle/Belle II].
3. *Search for CP violation in $\Xi_c^+ \rightarrow \Sigma^+ h^+ h^-$ and $\Lambda_c^+ \rightarrow p h^+ h^-$ decays*, Phys. Rev. D **113**, 032017 (2026) [Belle II].
4. *Observation of the singly Cabibbo-suppressed decays $\Xi_c^+ \rightarrow p K_S^0, \Lambda \pi^+, \Sigma^0 \pi^+$* , J. High Energy Phys. **03** (2025) 061 [Belle/Belle II].
5. *Measurement of branching fractions of $\Xi_c^0 \rightarrow \Xi^0 \pi^0, \Xi^0 \eta, \Xi^0 \eta'$* , J. High Energy Phys. **10** (2024) 045 [Belle/Belle II].
6. *Observation of $B^+ \rightarrow \Sigma_c(2455)^{++} \bar{\Xi}_c^-$ and $B^0 \rightarrow \Sigma_c(2455)^0 \bar{\Xi}_c^0$* , Phys. Rev. D **112**, L051101 (2025) [Belle/Belle II].